

## Lagrangian & Hamiltonian Approaches

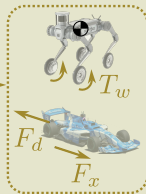
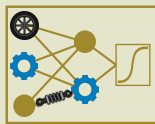


$$\mathcal{L} = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M}(\mathbf{q}) \dot{\mathbf{q}} - \mathcal{V}(\mathbf{q})$$

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{c}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{g}(\mathbf{q}) = \boldsymbol{\tau}$$

*Learning dynamics  
from variational principles*

## Model-Structured



*Physics-based layers,  
connections, constraints*

## NODE & Variational Integrator Networks



$$\frac{dq}{dt} = f_{\text{NN}}(\cdot)$$

$q_{t+1} \leftarrow$  variational integrator

*Learning dynamics via  
structure-preserving integration*

## Hybrid Physics-Learning

*Residual Learning*

Physics  
Model



*Subsystems  
Learning*

Physics

Physics



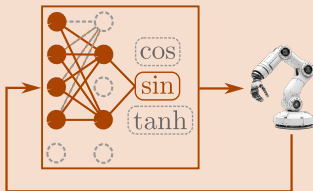
*Sensor  
Processing*



Physics



## Physics-Encoded Topology Learning



*Learning connectivity  
from physics priors*

## Physics-Encoded Neural Operators

